Week 15 - Northwind - DB_First

API:

Step 1: Install needed Packages In API :

```
NorthwindCatalog.Services   NorthwindCatalog.Tests   UnitTest1.cs

<Project Sdk="Microsoft.NET.Sdk.Web">

  <PropertyGroup>
    <TargetFramework>net8.0</TargetFramework>
    <Nullable>enable</Nullable>
    <ImplicitUsings>enable</ImplicitUsings>
  </PropertyGroup>

    <ItemGroup>
        <PackageReference Include="AutoMapper" Version="12.0.1" />
        <PackageReference Include="AutoMapper.Extensions.Microsoft.DependencyInjection" Version="12.0.1" />
        <PackageReference Include="Microsoft.VisualStudio.Web.CodeGeneration.Design" Version="8.0.23" />
        <PackageReference Include="Microsoft.EntityFrameworkCore.SqlServer" Version="8.0.1" />
        <PackageReference Include="Microsoft.EntityFrameworkCore.Tools" Version="8.0.23">
            <IncludeAssets>runtime; build; native; contentfiles; analyzers; buildtransitive</IncludeAssets>
            <PrivateAssets>all</PrivateAssets>
        </PackageReference>
        <PackageReference Include="Swashbuckle.AspNetCore" Version="6.6.2" />
    </ItemGroup>

</Project>

102 %   No issues found   Ln: 1, Ch: 1   MIXED   CRLF   UTF-8 with BOM
```

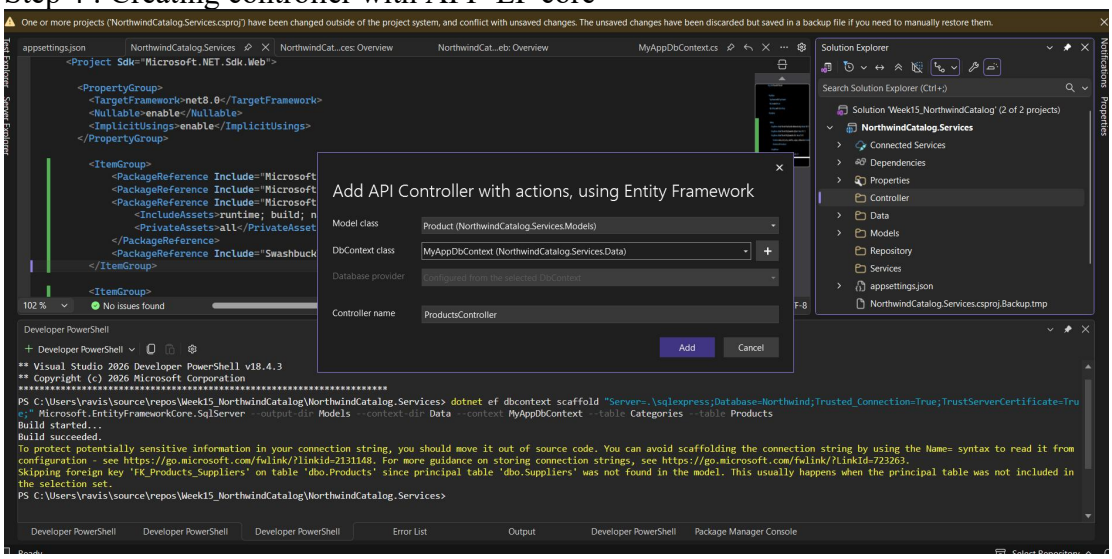
Step 2: Connection String Implementation

```
appsettings.json*   NorthwindCatalog.Services   NorthwindCat...ces: Overview   NorthwindCat...eb: Overview   MyAppDbContext.cs
Schema: https://www.schemastore.org/appsettings.json

{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
  "ConnectionStrings": {
    "DefaultConnection": "Server=.\\sqlexpress;Database=Northwind;Trusted_Connection=True;TrustServerCertificate=True;"
  }

}

102 %   No issues found   Ln: 10, Ch: 66   SPC   CRLF   UTF-8
```

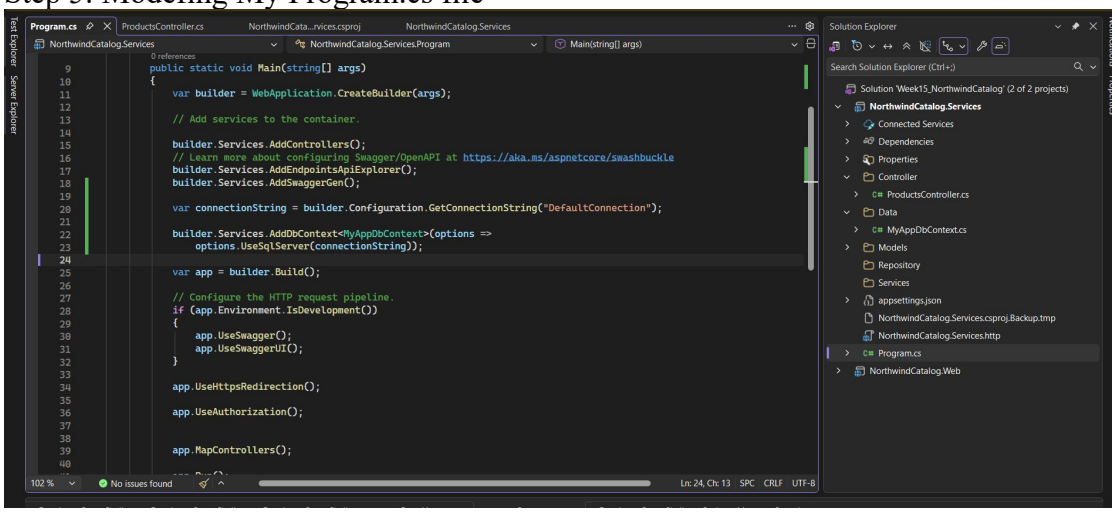
Step 3: Command to Get the Model and Dbcontext file from database

```
Developer PowerShell

** Visual Studio 2026 Developer PowerShell v18.4.3
** Copyright (c) 2026 Microsoft Corporation
**********************************************************************
PS C:\Users\ravis\source\repos\Week15_NorthwindCatalog\NorthwindCatalog.Services> dotnet ef dbcontext scaffold "Server=.\sqlexpress;Database=Northwind;Trusted_Connection=True;TrustServerCertificate=True;" Microsoft.EntityFrameworkCore.SqlServer --output-dir Models --context-dir Data --context MyAppDbContext --table Categories --table Products
Build started...
Build succeeded.
To protect potentially sensitive information in your connection string, you should move it out of source code. You can avoid scaffolding the connection string by using the Name= syntax to read it from configuration - see https://go.microsoft.com/fwlink/?linkid=2131148. For more guidance on storing connection strings, see https://go.microsoft.com/fwlink/?LinkId=723263.
Skipping foreign key 'FK_Products_Suppliers' on table 'dbo.Products' since principal table 'dbo.Suppliers' was not found in the model. This usually happens when the principal table was not included in the selection set.
```

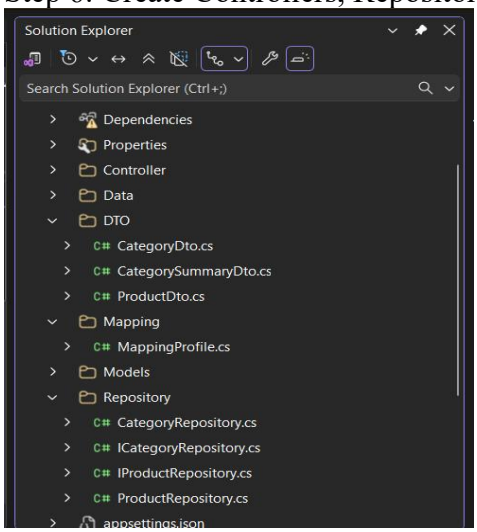
Step 4 : Creating controller with API- EF core



Step 5: Modefing My Program.cs file



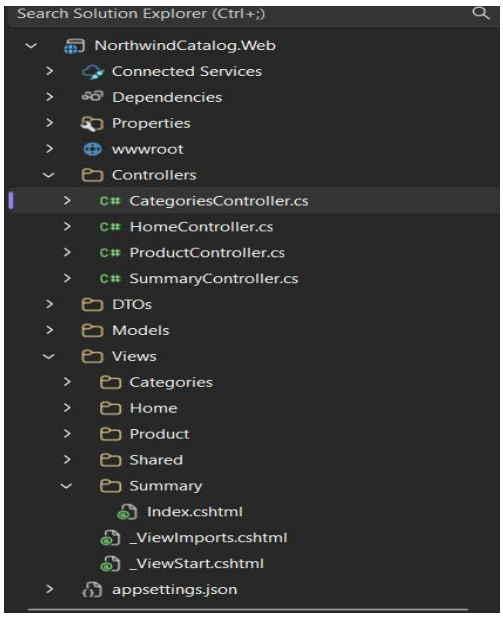
Step 6: Create Controllers, Repository and DTO



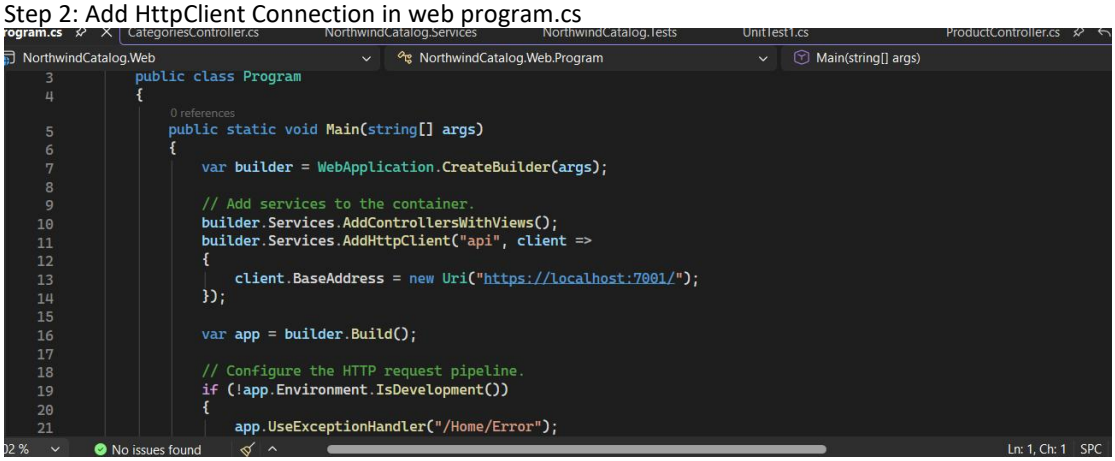
Step 7: Check Your API on Swagger

Web:

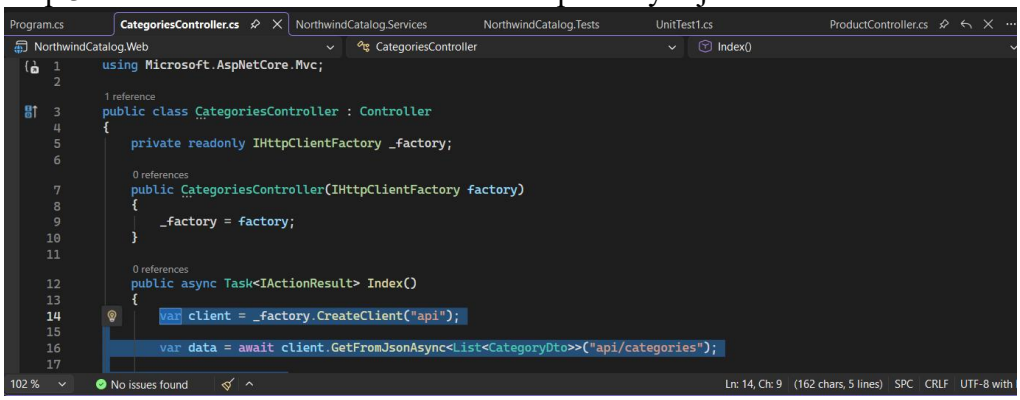
Step 1: File structure



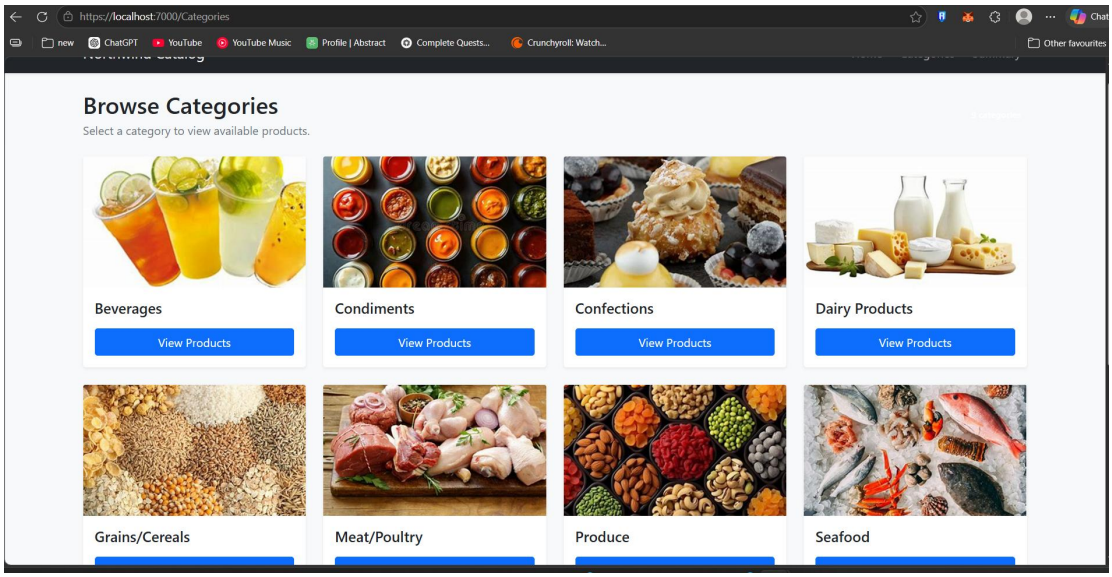
Step 2: Add HttpClient Connection in web program.cs

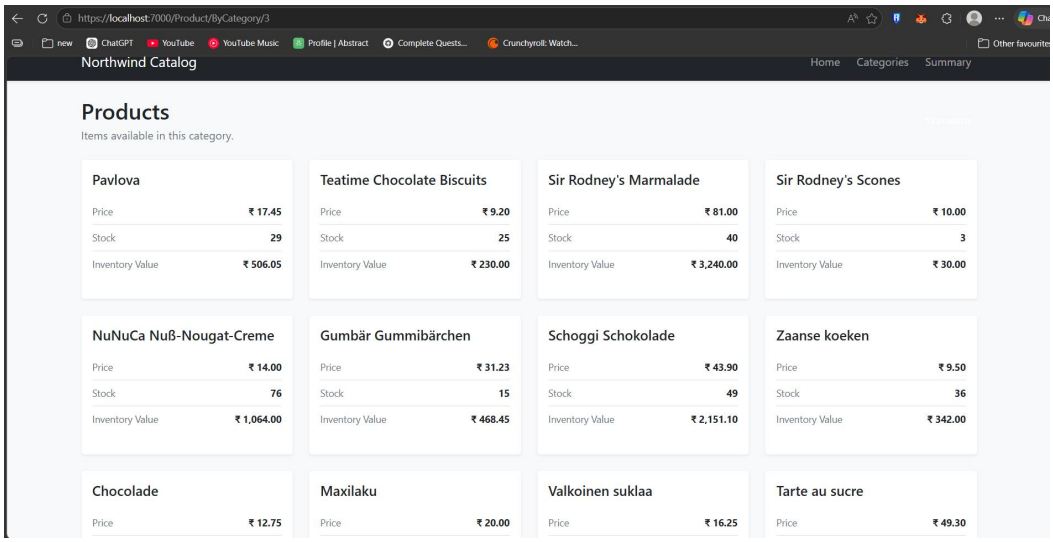


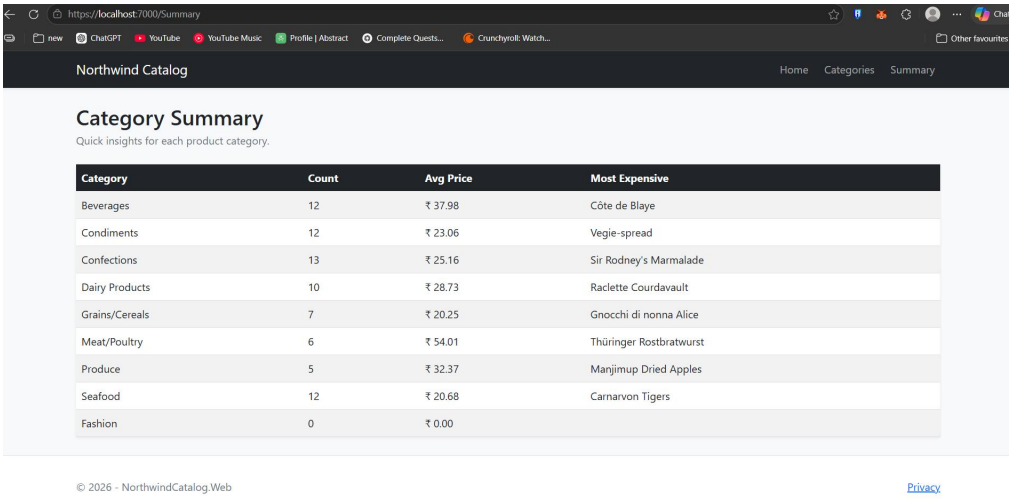
Step 3: Now Create Controllers and use Dependacy Injection



Step 4: Create Views for all of your Action Method
Step 5: use BootStrap To improve the UI/UX of your website Page



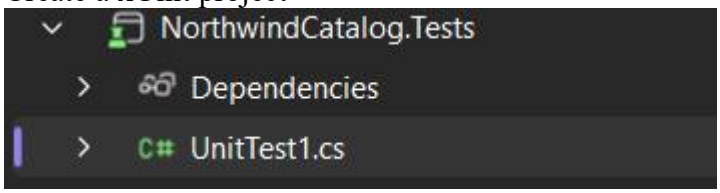




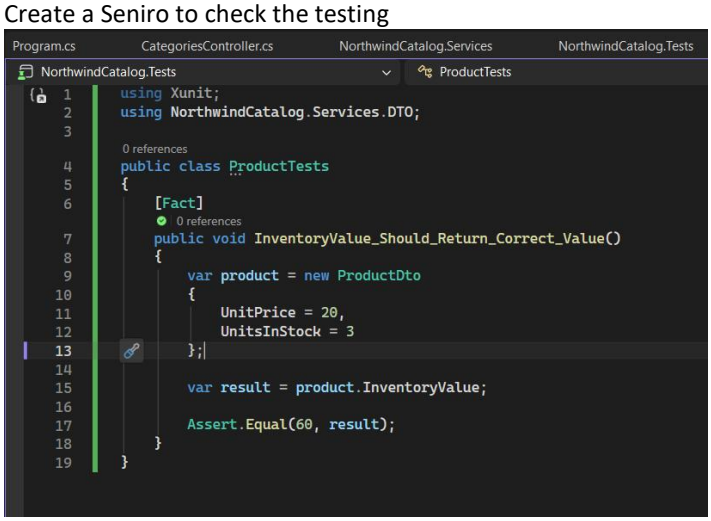
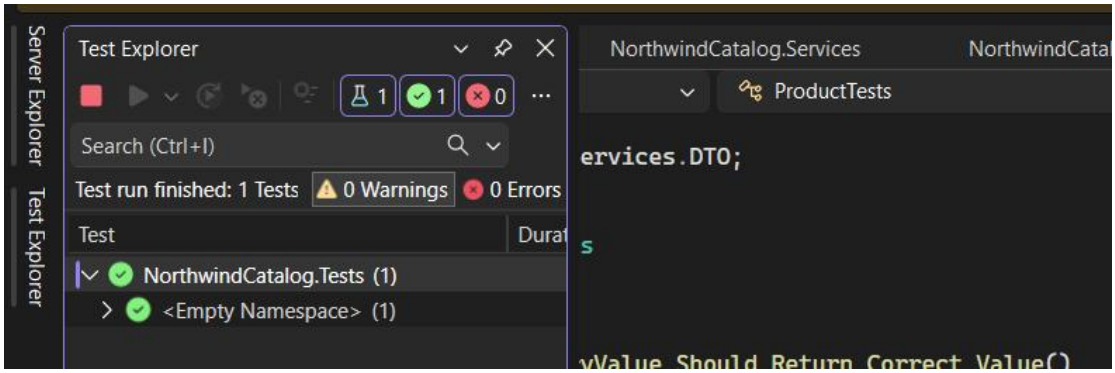
Category	Count	Avg Price	Most Expensive
Beverages	12	₹ 37.98	Côte de Blaye
Condiments	12	₹ 23.06	Vegie-spread
Confections	13	₹ 25.16	Sir Rodney's Marmalade
Dairy Products	10	₹ 28.73	Raclette Courdavault
Grains/Cereals	7	₹ 20.25	Gnocchi di nonna Alice
Meat/Poultry	6	₹ 54.01	Thüringer Rostbratwurst
Produce	5	₹ 32.37	Manjimup Dried Apples
Seafood	12	₹ 20.68	Carnarvon Tigers
Fashion	0	₹ 0.00	

Testing:

Create a xUnit project



Create a Seniro to check the testing



Check your testing